

RIT*: Riemannian Informed Trees for Cost-Adaptive Optimal Motion Planning

Muhayy Ud Din¹, Ahmed Nadar¹, Jan Rosell², and Irfan Hussain^{1*}

Abstract—We present Riemannian Informed Trees (RIT*), a planning framework that replaces Euclidean primitives in batch-informed search with their Riemannian counterparts. RIT* constructs a tighter, cost-consistent informed set, performs a nearest-neighbour search under an anisotropic distance metric, and evaluates edge costs efficiently via a cascading scheme. We further introduce a Collision-Adaptive Metric Refinement (CARM), which learns an obstacle-proximity cost field online from collision feedback, reducing the reliance on prior metric design in practical settings. Experiments across environments from 2-D to 14-D show that RIT* is competitive in low-dimensional and spatially constant-metric settings and produces substantially lower-cost solutions when the metric varies spatially in high-dimensional configuration spaces. Performance gains scale with anisotropy and dimension, reaching up to 13.0% improvement in median initial cost over BIT* in the 3-D anisotropic benchmark, up to 9.0% in median final cost over BIT* in 6-DOF manipulation, and 24.8-63.5% in a 14-DOF bimanual planning problem, where Euclidean-informed baselines degrade.

Index Terms—Motion planning, Riemannian geometry, asymptotic optimality, informed sampling, online metric learning.

I. INTRODUCTION

Sampling-based motion planning enables collision-free path computation in high-dimensional configuration spaces [1]. Asymptotically optimal (AO) planners such as RRT* [2] and PRM* converge to an optimal path, and later work improved convergence via *informed sampling*: once an initial solution is found, samples are drawn solely from the *informed set*, a prolate hyperspheroid in Euclidean space [3]. This idea underpins Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6].

These planners, however, select this set using Euclidean distance, which does not reflect the true cost structure in anisotropic environments. In addition, they typically rely on a predetermined cost structure rather than learning it from planning experience, reducing their effectiveness when the cost landscape is unknown or environment-dependent.

In this work, we address both limitations with RIT*. We replace Euclidean components with a Riemannian formulation, where a spatially varying metric tensor $G(x)$ captures the underlying cost structure. We further introduce a Collision-Adaptive Metric Refinement (CARM), which learns

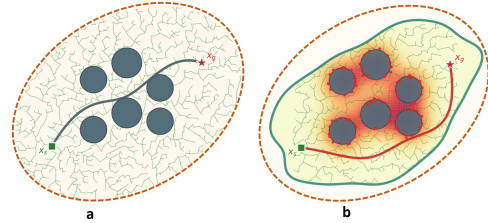


Fig. 1: Overview of RIT*. (a) BIT* samples from the Euclidean informed set I_E , ignoring anisotropic cost structure. (b) RIT* uses a cost-aligned Riemannian informed set; CARM (red shading) learns a cost field from collision feedback to guide paths away from obstacles.

a collision-induced conformal cost field while planning from collision feedback and adapts the planner accordingly (Fig. 1).

Contributions: The key contributions are:

- 1) *Riemannian informed planning*: RIT* replaces Euclidean components with Riemannian ones: the informed set aligns with the underlying cost structure, nearest neighbours use an anisotropic metric, and edge costs are computed via a cascading scheme, focusing sampling on relevant regions and improving efficiency.
- 2) *Online proximity-aware metric refinement via CARM*: CARM learns an obstacle-proximity cost field from collision feedback and applies it as a conformal scaling of a base metric, requiring no prior obstacle knowledge: given a base metric (e.g., joint-space inertia) it adds proximity-aware structure; without one, it defaults to Euclidean and learns from scratch.
- 3) *Experimental validation*: We evaluate RIT* from 2-D/3-D to high-dimensional tasks (6-DOF, 14-DOF), including a real UR10e arm, with a comprehensive benchmark against asymptotically optimal planners.

II. RELATED WORK

We review prior work on informed sampling-based planning, the use of Riemannian geometry in motion planning, and obstacle-aware cost design.

Euclidean informed sampling planners: Sampling-based planners have been improved by restricting search to an informed subset of the space. Gammell et al. [3] introduced direct sampling of a prolate hyperspheroid to accelerate RRT*. BIT* [4] combines batch sampling with informed sets and lazy edge evaluation, while AIT* and EIT* [5] extend this with asymmetric bidirectional search. APT* [6] shapes nearest-neighbour search using virtual forces and adapts batch size,

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and FIT* [7] improves early convergence through batch adaptation. Despite these advances, all define the informed set, nearest-neighbour search, and edge cost in *Euclidean* distance.

Riemannian geometry in motion planning. Riemannian geometry naturally models anisotropic costs, but its use is largely limited to local or optimisation methods. CHOMP [8] and STOMP [9] optimise trajectories under such costs, but are not sampling-based and lack AO guarantees. In sampling-based planning, Kim et al. [10] use Riemannian metrics for local steering, and Kyaw and Kelly [11] propose a Riemannian RRT* using retractions and natural gradients. However, these methods are incremental, do not construct informed sets, and do not address global sampling. Bi-AM-RRT* [12] similarly biases nearest-neighbour selection via a diffusion-based assisting metric, but does not integrate informed sampling or batch processing, its metric serves as a search heuristic for fast (re)planning in dynamic 2-D navigation rather than as the cost functional being optimised. The cost-space RRT literature, notably T-RRT on configuration-space costmaps [13], biases exploration through stochastic transition tests over a scalar cost field. These planners handle general costmaps but retain Euclidean distances, are not informed, and provide no asymptotic-optimality guarantees..

Obstacle-aware Riemannian metrics. Several works incorporate obstacle information into metric design but rely on prior knowledge or demonstrations: Klein et al. [14] design barrier-based metrics requiring full obstacle geometry, RMPflow [15] composes task-space metrics for reactive control, and Beik-Mohammadi et al. [16] learn Riemannian manifolds from demonstrations. None is integrated with sampling-based planning, and all require information beyond collision checking.

Key distinction: Prior Riemannian approaches use the metric for *local* operations (steering, trajectory optimisation, reactive control). RIT* instead integrates Riemannian geometry into the *global* informed sampling pipeline, replacing the informed set, nearest-neighbour search, edge cost evaluation, and rewiring of a batch-informed planner with Riemannian counterparts, and incorporates CARM, which learns a conformal scaling online from collision feedback, enabling cost-aware planning without prior environment knowledge.

III. PROBLEM FORMULATION

We consider optimal motion planning where the cost of motion is anisotropic and may vary across the space; such costs are naturally described by a Riemannian metric assigning a direction-dependent traversal cost at each configuration. Let $\mathcal{C} \subseteq \mathbb{R}^d$ denote a d -dimensional configuration space, and let $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ be the subset of collision-free configurations. The space is equipped with a smooth, positive-definite metric tensor field $G : \mathcal{C} \rightarrow \mathbb{S}_{++}^d$, where $\mathbb{S}_{++}^d$ denotes the set of symmetric positive-definite matrices. For any configuration $x \in \mathcal{C}$, the matrix $G(x)$ defines the local cost of moving through x , so that different directions can have different traversal costs.

Under this metric, the cost of a path $\sigma : [0, 1] \rightarrow \mathcal{C}$ is its Riemannian arc length,

$$c(\sigma) = \int_0^1 \sqrt{\dot{\sigma}(t)^\top G(\sigma(t)) \dot{\sigma}(t)} dt, \quad (1)$$

where $\dot{\sigma}(t)$ is the tangent vector of the path. The resulting *geodesic distance* $d_R(x, y)$ is defined as the minimal cost of any path between the two configurations x and y , i.e., $d_R(x, y) = \inf_{\sigma} c(\sigma)$ taken over all paths σ that connect x and y . In the special case where $G(x) = I_d$, this corresponds to the usual Euclidean distance.

Given a start state $x_s \in \mathcal{C}_{\text{free}}$ and a goal state $x_g \in \mathcal{C}_{\text{free}}$, our objective is to find a collision-free path of minimum cost:

$$\sigma^* = \arg \min_{\sigma : x_s \rightarrow x_g, \sigma \subset \mathcal{C}_{\text{free}}} c(\sigma), \quad (2)$$

A planner is *asymptotically optimal* (AO) if the cost c_n of its returned solution converges almost surely to the optimal cost c^* as the number of samples $n \rightarrow \infty$.

After a feasible solution with cost c_{best} is found, the search can be restricted to configurations that may still generate a better path. In Euclidean informed planning, this region is the *informed set*

$$I_E = \{x \in \mathcal{C} : \|x_s - x\|_2 + \|x - x_g\|_2 \leq c_{\text{best}}\}, \quad (3)$$

For anisotropic costs, however, Euclidean distance does not accurately reflect the true cost-to-come and cost-to-go. We therefore define the *Riemannian informed set* as

$$I_R = \{x \in \mathcal{C} : d_R(x_s, x) + d_R(x, x_g) \leq c_{\text{best}}\}. \quad (4)$$

This set contains configurations that can still improve the current solution under the Riemannian cost. When $G \succeq I_d$, $d_R(x, y) \geq \|x - y\|_2$ and hence $I_R \subseteq I_E$; more generally, under the bi-Lipschitz bounds of Sec. IV-F the two sets are equivalent up to constant-factor distortion, with I_R aligned to the cost structure.

Our goal is an AO informed planner operating natively under the Riemannian cost: sampling from I_R , selecting neighbours and evaluating edges via d_R , and, when the available metric is absent or does not encode obstacle proximity, refining it online from collision feedback.

Remark 1. *Numerical geodesic cost and admissible heuristic:* Since d_R has no closed form for general G , RIT* evaluates the arc-length of a straight-line edge $\gamma(t) = u + t\Delta$, $\Delta = v - u$, via n -point Gauss-Legendre (GL- n) quadrature,

$$c_R(u, v) \approx \sum_{k=1}^n w_k \sqrt{\Delta^\top G(u + t_k \Delta) \Delta}, \quad (5)$$

where $(t_k, w_k)_{k=1}^n$ are GL nodes and weights on $[0, 1]$ (default $n = 10$, configurable). For spatially constant metrics (e.g. $G = \text{diag}(w)$), eq. (5) is exact. The planning heuristic is $\hat{h}_R(v) = \sqrt{\Delta^\top G \Delta}$ for constant G , and $\hat{h}_R(v) = \sqrt{\lambda_{\min}} \|v - x_g\|_2$ for spatially varying G , where $\lambda_{\min} = \min_x \lambda_{\min}(G(x))$; admissibility follows from $d_R(x, y) \geq \sqrt{\lambda_{\min}} \|x - y\|_2$. We write c_R for the value (5) computes (the superscript $c_R^{(n)}$ is retained only in Lemma 1); since edges are straight lines, the tree-accumulated $\sum c_R$ converges to d_R under densification, so the implementation rewiring and pruning criteria are asymptotically equivalent to their d_R ideal forms.

Algorithm 1: RIT* planner

Input: $x_s, x_g, \mathcal{C}_{\text{free}}, G(\cdot), B, T, K_c, \sigma, \alpha$
Output: Optimal path σ^*

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1  $\mathcal{V} \leftarrow \{x_s\}; \mathcal{E} \leftarrow \emptyset; c_{\text{best}} \leftarrow \infty; \mathcal{C}_{\text{col}} \leftarrow \emptyset;$ 
2 for  $t = 1, \dots, T$  do
3    $X_{\text{new}} \leftarrow \text{RIEMANNIANINFSAMPLE}(B, c_{\text{best}}, G)$ 
4    $\mathcal{V} \leftarrow \mathcal{V} \cup X_{\text{new}}$ 
5    $\text{PRUNENODES}(\mathcal{V}, c_{\text{best}}, G)$ 
6   Compute  $r_n^R$  via (6)
7   foreach  $v \in \mathcal{V}$  in order of  $f(v)$  do
8      $N(v) \leftarrow \text{neighbours of } v \text{ within } r_n^R$ 
9     foreach  $u \in N(v)$  do
10       $(c_{uv}, p_{\text{col}}) \leftarrow \text{CASCADINGEDGE}(u, v, G)$ 
11       $\text{UPDATEANDREWIRE}(\mathcal{V}, \mathcal{E}, u, v, c_{uv}, r_n^R, G)$ 
12      if  $v = x_g$  and  $g(v) < c_{\text{best}}$  then
13         $c_{\text{best}} \leftarrow g(v)$ 
14      end
15      if  $p_{\text{col}} \neq \emptyset$  then
16         $\mathcal{C}_{\text{col}} \leftarrow \mathcal{C}_{\text{col}} \cup \{p_{\text{col}}\}$ 
17      end
18    end
19  end
20  if  $t \bmod K_c = 0$  and  $|\mathcal{C}_{\text{col}}| > 0$  then
21     $G \leftarrow \text{CARM\_UPDATE}(\mathcal{C}_{\text{col}}, G, \sigma, \alpha)$ 
22  end
23 end
24 return  $\sigma^*$ 

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IV. METHOD

Batch Informed Trees [4] is an asymptotically optimal planner that incrementally builds a tree from batches of samples drawn from an informed set; within each batch, vertices and edges are ordered by cost-to-come and heuristic estimates and evaluated lazily, balancing global exploration with goal-directed search.

RIT* extends this framework with three modifications and a learning component: i) the informed set is defined by a Riemannian metric, producing a cost-aligned sampling region; ii) nearest-neighbour search uses Riemannian distance, enabling connections along low-cost directions; iii) edge costs are approximated via a cascading quadrature scheme; and CARM learns a conformal scaling online from collision feedback.

Algorithm 1 summarises the procedure. The planner takes as input a start-goal pair (x_s, x_g) , free space $\mathcal{C}_{\text{free}}$, a metric tensor G (or I_d if unknown), batch size B , iteration budget T , and CARM parameters: update interval K_c , bandwidth σ , and inflation strength α . A metric cache $\mathcal{M}$ is precomputed on a grid for $O(1)$ metric lookups. Within each batch, vertices are processed in order of $f(v) = g(v) + \hat{h}_R(v)$ (line 7), where $g(v)$ is the Riemannian cost-to-come and $\hat{h}_R(v) = \bar{c}(v, x_g)$ is a cached Riemannian distance heuristic (Remark 1). Components Riemannian sampling (line 3), cascading evaluation (line 10), rewiring, pruning (lines 5, 10), and CARM updates (line 21) are detailed later.

A. Riemannian Informed Sampling

RIT* draws samples from the Riemannian informed set I_R (line 3 of algorithm 1), defined in (4). When $G \succeq I_d$, the Riemannian distance dominates the Euclidean one, $d_R(x, y) \geq \|x - y\|_2$, so $I_R \subseteq I_E$.

To draw samples from I_R efficiently, we apply a metric whitening transform. Let $\bar{G}$ be the mean of G at $N_{\text{seg}} = 10$ uniformly spaced points along the start-goal segment and $L = \text{chol}(\bar{G})$; both are computed once per planner invocation (or per CARM update) and reused, so the per-sample cost

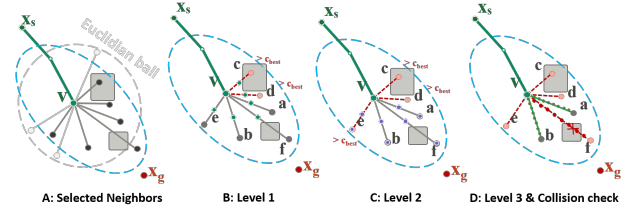


Fig. 2: Riemannian NN selection and cascading edge evaluation in RIT*. (a) The Riemannian ball expands along low-cost directions vs. the Euclidean ball. (b-d) Cascading filter: most edges are rejected early, only survivors receive full-precision evaluation.

is a single $L^{-\top}$ multiplication. The change of coordinates $\tilde{x} = L^{-\top}x$ maps I_R to an approximate prolate hyperspheroid, where the direct sampling of [3] applies. Samples are mapped back via $x = L^{-\top}\tilde{x}$. This transformation approximates the Riemannian informed set, with accuracy governed by the smoothness of G . Empirically, the median relative volume mismatch $|\text{vol}(L^{-\top}\mathcal{E}) - \text{vol}(I_R^{\text{true}})|/\text{vol}(I_R^{\text{true}})$ is below 1% on our 3-D, 6-D, and 14-D benchmarks and below 3% under mild spatial metric variation; in a controlled sweep with $G(x) = \text{diag}(1, 3, 6)(1 + \beta\|x - x_c\|^2)$, $\beta \in \{0, 0.5, 1, 2\}$, median mismatch grows smoothly from $< 0.5\%$ ($\beta = 0$) to 2.7% ($\beta = 1$) and 6.1% ($\beta = 2$), i.e. the whitening degrades gracefully with $\|\nabla G\|$.

B. Riemannian Nearest-Neighbour Search

Let r_n^R be the neighbour connection radius for n samples; RIT* uses the Riemannian ball $\{u : d_R(u, v) \leq r_n^R\}$, an ellipsoid aligned with the local metric that expands along low-cost directions and contracts along high-cost ones, matching the traversal cost structure. The connection radius is

$$r_n^R = \gamma_R \left(\frac{\log n}{n} \right)^{1/d}, \quad \gamma_R = 2 \left(1 + \frac{1}{d} \right)^{1/d} \left(\frac{\mu_R(I_R)}{\zeta_d} \right)^{1/d}, \quad (6)$$

where $\mu_R(I_R)$ is the Riemannian volume of I_R and ζ_d is the volume of the unit d -ball. Because $\mu_R(I_R) < \mu(I_E)$, this radius is typically smaller than its Euclidean counterpart, reducing the number of candidate edges. In practice we evaluate $\mu_R(I_R) \approx \det(\bar{G})^{1/2} \mu(I_E)$, the Jacobian change-of-variables under whitening (Sec. IV-A); this is exact for spatially constant G and incurs an error of $O(\|\nabla G\|_{\infty} \text{diam}(I_R))$ otherwise, quantified empirically in Sec. V-A.

C. Cascading Edge Evaluation

Evaluating the Riemannian edge cost requires numerical integration of (1) with multiple metric-tensor evaluations. Since most candidate edges cannot improve the current solution, RIT* screens them through a three-level cascade of increasing accuracy and cost, rejecting unpromising edges early (line 10 of algorithm 1).

For a candidate edge from parent u to vertex v , let $\Delta = v - u$ and $m = (u+v)/2$. An edge is discarded whenever $g(u) + \bar{c}(u, v) > c_{\text{best}}$, where $g(u)$ is the cost-to-come from x_s to u along the current tree.

L1: Midpoint check: A single cached metric lookup at the midpoint $m = (u + v)/2$ gives the fast estimate

$$\hat{c}_1 = \bar{c}(u, v) = \sqrt{\Delta^\top G(m) \Delta}. \quad (7)$$

This L1 estimate serves *only* as a rapid first-pass filter; the exact arc-length is computed at L3 via the Gauss-Legendre quadrature (Remark 1). L_1 rejects the majority of candidates across our 2-D to 14-D benchmarks (78-85%; 80% in the 6-D UR10 environment).

L2: Simpson estimate: Surviving edges are re-evaluated with Simpson's rule:

$$\hat{c}_2 = \frac{\|\Delta\|}{6} \left(\sqrt{\Delta^\top G(u) \Delta} + 4\sqrt{\Delta^\top G(m) \Delta} + \sqrt{\Delta^\top G(v) \Delta} \right), \quad (8)$$

which captures metric variation along the edge more accurately. This level rejects approximately 75% of the edges that passed L1.

L3: Full quadrature and collision check: Only the roughly 5% of edges surviving both levels receive 10-point Gauss-Legendre integration followed by collision checking. Fig. 2 illustrates the cascade: of seven candidate neighbours, two are rejected at L1, one at L2, and one at L3 due to collision, leaving three accepted edges.

Unlike the L3 quadrature (Lemma 1), the L1/L2 estimates (7)-(8) are not lower bounds on the true edge cost and could, in principle, discard an improving edge when G varies sharply along it. The cascade thresholds are therefore deflated: an edge is rejected at L1/L2 only if $g(u) + \eta_e^{-1} \hat{c} > c_{\text{best}}$, where $\eta_e = \sqrt{\lambda_{\text{max}}^e / \lambda_{\text{min}}^e}$ is computed from cached metric eigenvalue bounds over the cells traversed by the edge. Since every integrand sample of (1), and hence $\hat{c}_1$, $\hat{c}_2$, and $c_R(u, v)$, lies in $[\sqrt{\lambda_{\text{min}}^e} \|\Delta\|, \sqrt{\lambda_{\text{max}}^e} \|\Delta\|]$, we have $\eta_e^{-1} \hat{c} \leq \sqrt{\lambda_{\text{min}}^e} \|\Delta\| \leq c_R(u, v)$: thus, no improving edge is ever falsely rejected. With the bounded edge lengths of Table I and smooth G , $\eta_e \rightarrow 1$ as $\|\Delta\| \rightarrow 0$, so the deflation is mild in practice and vanishes under densification; the quoted rejection rates are reported with this safeguard in place.

D. Tree Rewiring and Pruning

Rewiring (line 11 of algorithm 1): When a new vertex v is added to the tree, it may provide a cheaper path to existing nearby vertices. For each neighbour u within the Riemannian radius r_n^R , RIT* checks whether the path through v is cheaper than the current parent of u :

$$g(v) + c_R(v, u) < g(u), \quad (9)$$

where $g(\cdot)$ is the Riemannian cost-to-come from x_s along the current tree and $c_R(v, u)$ is the Riemannian edge cost (1). If the condition holds, u is reconnected as a child of v and the cost-to-come of all descendants is updated. Because the comparison uses c_R rather than Euclidean distance, rewiring respects the anisotropic cost structure and discovers connections that Euclidean planners would overlook.

Pruning (line 5 of algorithm 1): After each batch, RIT* removes vertices that can no longer contribute to an improved solution. A vertex x is pruned if its best possible cost-to-come plus cost-to-go exceeds the current best solution cost:

$$d_R(x_s, x) + d_R(x, x_g) > c_{\text{best}}. \quad (10)$$

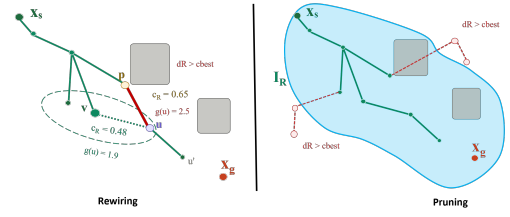


Fig. 3: Tree rewiring and pruning in RIT*. (a) Rewiring switches a vertex's parent when a lower-cost Riemannian path is found. (b) Pruning removes vertices outside the Riemannian informed set.

In practice, d_R in (10) is evaluated from the accumulated L3 edge costs c_R already computed during tree growth; by Remark 1 and Lemma 1, $c_R \rightarrow d_R$ under densification, preserving asymptotic optimality (Theorem 3). Pruning reduces the tree size and nearest-neighbour search cost, and when CARM updates the metric (line 21 of algorithm 1), it forces re-evaluation of vertex membership under the tightened I_R , ensuring the tree adapts to the evolving cost landscape (Fig. 3).

E. Collision-Adaptive Riemannian Metric (CARM)

CARM is an online *refinement* of an initial base metric: the planner starts from a base metric G , which defaults to the identity I_d when no prior metric is available, and CARM refines it from collision feedback. The metric G appearing as an input to algorithm 2 is thus the current (possibly default) metric being refined, not a prerequisite. CARM takes four inputs: a set of collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$ collected from failed edge checks, the current metric G , a kernel bandwidth σ that controls spatial resolution, and an inflation strength α that controls how aggressively traversal cost increases near obstacles.

Failed edge checks naturally reveal points near obstacle boundaries; as they accumulate, they form a sparse map of regions where traversal should be expensive. CARM converts this map into a density field via a Gaussian kernel density estimate (KDE),

$$\hat{f}(x) = \frac{1}{N} \sum_{i=1}^N \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right), \quad (11)$$

and applies it as a conformal scaling of the base metric (lines 1-3 of algorithm 2):

$$G_{\text{CARM}}(x) = s(x) G(x), \quad s(x) = 1 + \alpha \hat{f}(x). \quad (12)$$

Because $s(x)$ is scalar, G_{CARM} is a *conformal* refinement: CARM learns an obstacle-proximity inflation field, not an anisotropic tensor structure. Directional cost variation, when present, must come from the base metric $G(x)$; this is appropriate for collision-based learning because collisions carry positional information but not directional information. Once returned, G_{CARM} overwrites G in the main loop (line 21 of algorithm 1), so all subsequent sampling, edge evaluation and pruning use the refined metric. Since $G_{\text{CARM}}(x) \succeq G(x)$, traversal cost increases and the informed region contracts, preserving completeness and asymptotic optimality (Corollary 4). After each update the algorithm recomputes the metric cache, shrinks I_R , and prunes vertices outside the tightened

Algorithm 2: CARM_UPDATE($\mathcal{C}_{\text{col}}, G, \sigma, \alpha$)

Input: Collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$, current metric G , bandwidth σ , strength α

Output: Updated metric G_{CARM}

- 1 Build KDE: $\hat{f}(x) \leftarrow N^{-1} \sum_{i=1}^N \exp(-\|x - c_i\|^2 / (2\sigma^2))$
- 2 Compute scale field: $s(x) \leftarrow 1 + \alpha \hat{f}(x)$
- 3 Update metric: $G_{\text{CARM}}(x) \leftarrow s(x) G(x)$
- 4 Recompute metric cache at grid points
- 5 Recompute I_R with G_{CARM}
- 6 **return** G_{CARM}

TABLE I: Experimental environment summary. d : dimension; κ : metric condition number; ℓ_{max} : max edge length; n_c : collision checks per edge.

Environment	d	κ	ℓ_{max}	n_c
2-D Random World	2	1	0.05	10
3-D Diagonal	3	≈ 6	0.05	10
6-D UR10	6	≈ 12	0.10 rad	10
14-D Tiago Bimanual	14	≈ 18	0.10 rad	10

informed set (lines 4-6 of algorithm 2), creating a feedback loop: collisions refine the metric, the refined metric focuses sampling, and focused sampling generates more informative collisions.

CARM is triggered every K_c iterations (line 21 of algorithm 1) when the collision buffer is non-empty. As fig. 4 shows, the KDE field stabilises after ~ 20 iterations (~ 400 collision points in 2-D). We use $\sigma = 0.1$, $\alpha = 10$, and $K_c = 15$ throughout; Sec. V-E reports sensitivity to these choices.

F. Theoretical Analysis

RIT* preserves the asymptotic guarantees of BIT* [4]; we sketch the modifications needed under a Riemannian metric and CARM.

Assumption 1. G satisfies $\lambda_{\min} I_d \preceq G(x) \preceq \lambda_{\max} I_d$ on $\mathcal{C}_{\text{free}}$ with $0 < \lambda_{\min} \leq \lambda_{\max} < \infty$ and $\|\nabla G\|_{\infty} < \infty$; the optimal path σ^* has cost c^* and strong δ -clearance. For inertia metrics $G(q) = M(q)$ this holds when the configuration bounds exclude kinematic singularities; reported $\kappa = \lambda_{\max}/\lambda_{\min}$ is the maximum over 10^4 grid samples.

Bounding (1) by the extreme eigenvalues gives the bi-Lipschitz equivalence.

$$\sqrt{\lambda_{\min}} \|x - y\|_2 \leq d_R(x, y) \leq \sqrt{\lambda_{\max}} \|x - y\|_2, \quad (13)$$

Lemma 1 (Convergence of GL- n arc-length quadrature). *For any $u, v \in \mathcal{C}_{\text{free}}$ and smooth metric G , the n -point Gauss-Legendre estimate $c_R^{(n)}(u, v)$ defined in (5) satisfies*

$$|c_R^{(n)}(u, v) - c_R(u, v)| = O(\|v - u\|^{2n+1}), \quad (14)$$

where $c_R(u, v)$ is the true arc-length of the straight-line edge and the $O(\cdot)$ constant depends on the $(2n)$ -th derivative of $t \mapsto \sqrt{\Delta^\top G(u + t\Delta)} \Delta$ along the edge. The GL- n rule is exact for polynomials of degree $\leq 2n - 1$; for default $n = 10$ this yields a 20th-order method. For spatially constant metrics, the integrand is constant and $c_R^{(n)} = c_R$ exactly for any $n \geq 1$. Hence $c_R^{(n)} \rightarrow c_R$ as $\|v - u\| \rightarrow 0$ at rate $O(\|v - u\|^{2n+1})$, so the L3-evaluated pruning and rewiring are asymptotically exact, preserving Theorem 3.

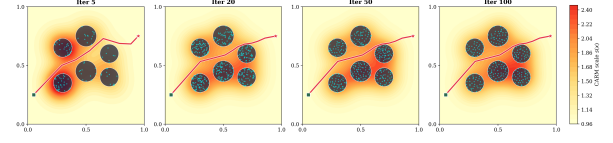


Fig. 4: CARM metric evolution: as collision points accumulate, the learned cost field refines, tightening the informed set and improving sample allocation.

Theorem 2 (Probabilistic completeness). *RIT* finds a feasible solution with probability 1 as $n \rightarrow \infty$.*

Proof. Discretize σ^* into waypoints $w_0, \dots, w_K$ with $\|w_k - w_{k+1}\|_2 < \delta/2$; each is in the fixed set $I_R^+ = \{x : d_R(x_s, x) + d_R(x, x_g) \leq c^* + \varepsilon\}$, which has a positive measure and is contained in I_R . Uniform sampling on I_R almost surely hits every $\delta/4$ -ball; by (13) the nearest samples satisfy $d_R(s_k, s_{k+1}) < \sqrt{\lambda_{\max}} \delta$, which falls below the connection radius $r_n^R = \Theta((\log n/n)^{1/d})$ for sufficiently large n , producing a collision-free chain $s_0 \rightarrow \dots \rightarrow s_K$ almost surely. $\square$

Theorem 3 (Asymptotic optimality). $P(\lim_{n \rightarrow \infty} c_n^{\text{RIT}^*} = c^*) = 1$.

Proof. We verify the Karaman-Frazzoli conditions [2]. By (13), a Riemannian ball of radius r_n^R contains a Euclidean ball of radius $r_n^R/\sqrt{\lambda_{\max}}$, so RIT*'s edge set is a superset of BIT*'s at $\tilde{r}_n = r_n^R/\sqrt{\lambda_{\max}}$ and inherits its neighbour-graph properties. Using (6), $\mu_R(I_R) \geq \lambda_{\min}^{d/2} \mu(I_R) > 0$, so the connection-radius constant $\gamma_R \geq \lambda_{\min}^{1/2} \gamma_{\text{KF}}$ (where γ_{KF} is the Karaman-Frazzoli lower bound on γ in (6)) exceeds the Karaman-Frazzoli threshold. The remaining steps (dense approximation of σ^* , cost convergence, optimal rewiring) follow [4] with d_R replacing $\|\cdot\|_2$ and (13) together with the GL- n convergence bound of Lemma 1 bounding the approximation error. $\square$

Corollary 4 (CARM preserves guarantees). *With CARM, $1 \leq s(x) \leq 1 + \alpha$, so the scaled metric satisfies Assumption 1 with $(\lambda_{\min}, (1 + \alpha)\lambda_{\max})$. Updates terminate at iteration t_0 when $\|\hat{f}_t - \hat{f}_{t-1}\|_{\infty} < \varepsilon_0$ ($\varepsilon_0 = 10^{-3}$); since at most B collision points enter per K_c -window, the per-update change is $O(B/N_t) \rightarrow 0$, so $t_0 < \infty$ almost surely. For $t \geq t_0$, the metric is frozen at G_{∞} and Theorems 2 and 3 apply with its constants. Since $G_{\infty} \succeq G$ implies $d_R^{G_{\infty}} \geq d_R^G$ and hence $I_R^{G_{\infty}} \subseteq I_R^G$, the pruning rule (10) applied under G_{∞} targets the G_{∞} -optimal solution, which is the planner's objective after CARM refinement.*

V. EXPERIMENTS

A. Experimental Setup

We evaluated RIT* in environments that span 2-D, 3-D, 6-D and 14-D configuration spaces (Table I, Fig. 5), designed to assess both the Riemannian informed-set construction and the CARM adaptation mechanism¹.

In low-dimensional spaces ($\mathbb{R}^2$ and $\mathbb{R}^3$), we consider two geometric benchmarks. The *Random World* (2-D) consists

¹Code and scripts to reproduce all reported results are provided as supplementary material and will be released publicly.

TABLE II: Benchmark Evaluations. Per environment: $t_{\text{init}}^{\min}$, $t_{\text{init}}^{\text{med}}$ (min/median time to first solution, s), $c_{\text{init}}^{\text{med}}$ (median initial cost), and $c_{\text{fin}}^{\text{med}}$ (median final cost). **Bold**: best per column.

Planner	2-D Random World ($\mathbb{R}^2$)				3-D Diagonal ($\mathbb{R}^3$)				6-D UR10 ($\mathbb{R}^6$)				14-D Tiago Biman. ($\mathbb{R}^{14}$)				Succ. (%)
	t_{min}	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	t_{min}	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	t_{min}	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	t_{min}	t_{med}	$c_{\text{init}}^{\text{med}}$	$c_{\text{fin}}^{\text{med}}$	
RIT*	0.013	0.020	0.9578	0.9135	0.021	0.040	2.4923	2.1608	0.0601	0.0915	1.9143	1.4143	0.8135	3.4255	16.5747	8.5778	100
BIT*	0.0183	0.0308	0.9698	0.9108	0.0476	0.0698	2.8167	2.2027	0.0651	0.0820	1.9415	1.5415	0.7657	0.8834	18.5069	10.7092	100
Inf. RRT*	0.0129	0.0289	0.9946	0.9165	0.0312	0.0464	2.8263	2.2278	0.1390	0.2114	1.9541	1.6975	1.5915	2.2198	19.9509	13.5773	100
AIT*	0.0111	0.0217	1.0475	0.9116	0.0236	0.0433	2.9950	2.2563	0.0697	0.1159	2.7794	1.4588	0.8070	1.4469	18.8514	11.0986	100
EIT*	0.0106	0.0162	1.0285	0.9124	0.0247	0.0413	2.8736	2.2449	0.0705	0.1397	2.7929	1.4530	0.8095	1.4452	19.8802	11.4415	100
APT*	0.0127	0.0189	1.0946	0.9225	0.0312	0.0465	2.7233	2.4436	0.1391	0.2126	2.9256	1.7536	1.6003	2.2284	19.8939	14.0232	100

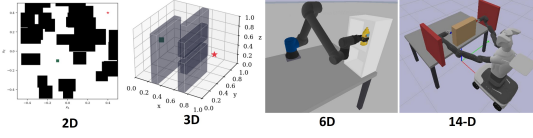


Fig. 5: Benchmark environments from 2-D to 14-D, covering geometric setups and manipulation tasks for RIT* evaluation.

of a square populated with 30 obstacles under a Euclidean metric ($\kappa = 1$), providing an isotropic baseline. The *Diagonal* environment (3-D) is a unit cube with obstacles aligned with the axis that form a narrow passage, equipped with a diagonal-anisotropic metric $G = \text{diag}(1, 3, 6)$ ($\kappa \approx 6$) to evaluate direction-dependent costs.

In higher-dimensional manipulation tasks ($\mathbb{R}^6$, $\mathbb{R}^{14}$), we evaluate scalability under articulated robot dynamics. In $\mathbb{R}^6$, a UR10e arm with a Robotiq 85 gripper in PyBullet [17] uses a joint-space inertia metric $G(q) = M(q)$ ($\kappa \approx 12$); the task (*UR10 Drill*) retrieves a drill from a cluttered shelf. In $\mathbb{R}^{14}$, a *Tiago Pro Bimanual* task with two 7-DOF arms performs a coordinated pre-grasp motion toward a tall box with obstacle pillars blocking direct paths, under a bimanual inertia metric ($\kappa \approx 18$), our most challenging setting in dimensionality and anisotropy.

We compare RIT* with five asymptotically optimal anytime planners: Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6]. All planners are implemented in a common Python framework with shared collision checks, random seeds, and environment interfaces; each is provided with the same metric tensor G and evaluates edge costs using the same Riemannian integral (1), ensuring a common objective, and the cascading evaluation of Sec. IV-C is applied to all planners, so time differences are attributable to informed-set geometry and connectivity rather than integration cost. Baselines retain their original Euclidean informed sets and nearest-neighbour radii, isolating the effect of the Riemannian replacements. All planners use batch size $B = 100$, maximum iterations $T = 200$, a per-trial timeout of 200s, and base seed 60; edge parameters follow Table I. All path costs in Tables II-IV are evaluated *a posteriori* under the fixed base metric G of each environment (the oracle metric in 6-D/14-D), never under G_{CARM} : CARM only shapes where the planner samples, connects, and prunes, so metric inflation cannot bias the reported costs. The oracle-vs-CARM comparison in Table IV therefore directly measures the base-metric suboptimality introduced by CARM’s conservativeness, including any bias from spurious collision clusters, bounding it at $\leq 1.0\%$; moreover, since the KDE (11) is normalised by N , isolated spurious clusters receive vanishing weight as collision evidence accumulates. We

do not compare against the metric-adaptive Bi-AM-RRT* [12]: its diffusion-based assisting metric accelerates search and replanning in dynamic 2-D navigation but is decoupled from the optimisation objective, so under the cost (1) it optimises a different objective, making a path-cost comparison inappropriate. An efficiency-focused comparison in dynamic settings, where Bi-AM-RRT* is designed to excel, is left to future work. Runtime profiling on the 14-D benchmark shows collision checking dominates ($\sim 61\%$ of wall-clock time), followed by nearest-neighbour search (21%), metric evaluation/cache lookup (11%), and CARM updates (3%); the cascade reduces full quadrature and collision checks to $\sim 5\%$ of candidates.

For RIT*, we use a diagonal approximation of the metric, $G(q) = \text{diag}(M(q))$, evaluated via trilinear interpolation from a precomputed cache on a res^d grid with $O(1)$ lookup ($\text{RES} = 32$ in 2-D/3-D, $\text{RES} = 3$ in 6-D/14-D). This trade-off preserves the dominant per-joint inertia scaling that drives the reported anisotropy ($\kappa \approx 12$ in 6-D, $\kappa \approx 18$ in 14-D) while avoiding the $O(d^2)$ cache footprint and full $M(q)$ evaluations at every L1 lookup. The coarse 6-D/14-D resolution is justified by the smooth variation of $M(q)$ within the planning bounds (relative interpolation error $< 4\%$ at 10^4 random configurations) and because the cache feeds only the L_1 filter; L_3 re-samples $G(q)$ on the fly.

Each experiment uses $N = 50$ trials with a 200s timeout. We report wall-clock time and Riemannian path cost; *init/final* denote the first and terminal solutions, and **bold** marks the best per column. The volume approximation of Sec. IV-B incurs $< 1\%$ error on our benchmarks and 2.7% at $\beta = 1$ in the controlled sweep of Sec. IV-A, consistent with the bound.

B. Low-Dimensional Benchmarks (2-D, 3-D)

In isotropic 2-D ($\kappa = 1$), RIT* attains the lowest initial cost (0.958), with gaps to baselines of 1.3-14.3%; Euclidean and Riemannian informed sets coincide here, so the small lead reflects only the tighter NN ellipsoid. At convergence all planners cluster within 1.3% ($c_{\text{fin}}^{\text{med}} \in [0.911, 0.923]$), BIT* marginally lowest at 0.911 vs. RIT* 0.914), confirming the predicted equivalence under isotropy. In 3-D ($\kappa \approx 6$), RIT* attains the lowest cost (2.49) and is fastest in median time to a first solution (0.040s, narrowly ahead of EIT* at 0.041s), outperforming APT* (2.72, +9.3%), BIT* (2.82, +13.0%), IRRT* (2.83, +13.4%), EIT* (2.87, +15.3%), and AIT* (2.99, +20.1%). The 3-D Diagonal metric is spatially constant, so the mean-metric whitening of Sec. IV-A is exact and I_R is a uniformly rescaled I_E ; the gain over BIT* here stems from the tighter nearest-neighbour ellipsoid, with informed-set tightening becoming dominant when the metric varies spatially (Sec. V-C). The lead persists at convergence: RIT* attains

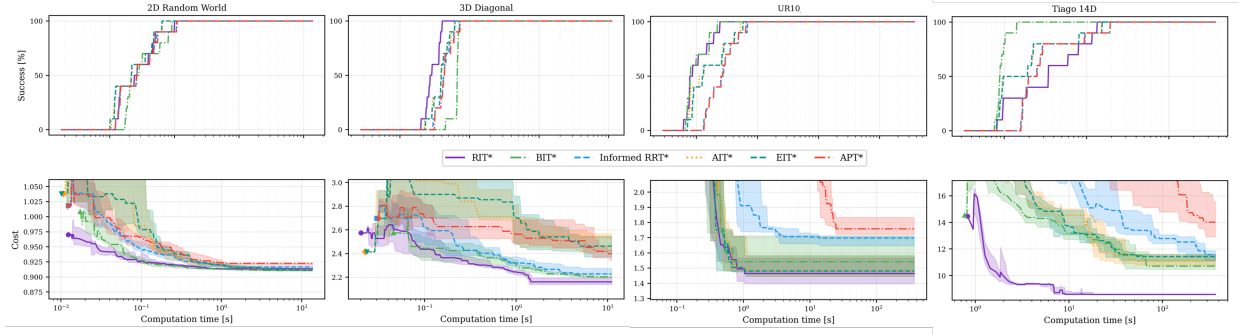


Fig. 6: Success rate (top) and Riemannian path cost (bottom) vs. time across all benchmarks; median over trials with shaded spread. RIT* converges faster and to lower cost, with the largest gains at 6-D/14-D.

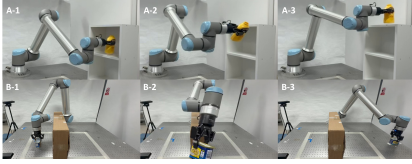


Fig. 7: UR10e real-robot execution of RIT*-planned trajectories. The planner produces smooth, collision-free motions that transfer directly from simulation to hardware.

the lowest $c_{\text{fin}}^{\text{med}} = 2.16$, with gaps of +1.9-13.1% across baselines.

C. High-Dimensional Benchmarks (6-D, 14-D)

In the 6-D UR10 task ($\kappa \approx 12$), RIT* achieves the lowest cost across all baselines (1.91) while maintaining complete success (100%); RIT* improves on BIT* (1.94, +1.4%) and IRRT* (1.95, +2.1%); AIT* (2.78, +45.2%), EIT* (2.79, +45.9%), and APT* (2.93, +52.8%) trail further. At convergence RIT* retains the lead ($c_{\text{fin}}^{\text{med}}$: RIT* 1.414, EIT* 1.453, AIT* 1.459; spread 3.2%), with BIT* (+9.0%), IRRT* (+20.0%), and APT* (+24.0%) trailing; the Riemannian informed set tightens both early progress and final convergence in this regime. In the 14-D Tiago task ($\kappa \approx 18$), RIT* attains the lowest cost (8.58), outperforming BIT* (10.71, +24.8%), AIT* (11.10, +29.4%), and EIT* (11.44, +33.4%), with IRRT* (13.58, +58.3%) and APT* (14.02, +63.5%) trailing further. The same ordering holds for $c_{\text{fin}}^{\text{med}}$, where RIT*'s lead remains the largest across all environments, indicating that informed-set tightening, not just early connectivity, drives the gain at high dimension and anisotropy. Performance gains grow with both anisotropy and dimensionality: up to +14.3% (2-D), +20.1% (3-D), up to +24.0% in 6-D, and +24.8–63.5% (14-D, across all five baselines). RIT*'s median time to a *first* solution in 14-D (3.43 s) is about four times that of BIT* (0.88 s). This overhead has two sources: the smaller Riemannian connection radius (6), since $\mu_R(I_R) < \mu(I_E)$ results in fewer candidate edges per batch and consequently delays first connectivity in 14 dimensions, and the additional cost of per-edge metric evaluations. This initial overhead is offset after the first solution: RIT* converges to a 24.8% lower final cost than BIT* (Fig. 6). Where first-solution latency is critical, a hybrid schedule that uses the Euclidean connection radius until an initial solution is found could reduce this overhead without affecting the asymptotic guarantees.

TABLE III: Ablation study: mean path cost ($\pm$ std) over 50 trials. **Bold**: best practical variant with complete success (oracle excluded). “w/o CARM” uses the oracle metric.

Variant	2-D	3-D	6-D	14-D
RIT* (full)	0.921 $\pm$ 0.007	2.196 $\pm$ 0.027	1.489 $\pm$ 0.082	8.638 $\pm$ 0.144
w/o Riem. samp.	0.916 $\pm$ 0.002	2.388 $\pm$ 0.182	1.459 $\pm$ 0.075	9.226 $\pm$ 0.935
w/o cascading	0.920 $\pm$ 0.008	2.195 $\pm$ 0.017	1.489 $\pm$ 0.082	8.638 $\pm$ 0.144
w/o CARM	0.913 $\pm$ 0.003	2.174 $\pm$ 0.033	1.484 $\pm$ 0.086	8.586 $\pm$ 0.012

Real robot execution: We additionally demonstrate execution of RIT*-planned trajectories on a UR10e platform (Fig. 7); a full hardware comparison is left to future work. The trajectories are smooth, collision-free, and preferentially use lower-inertia joints (wrist over shoulder/elbow), consistent with simulation. Across 20 hardware runs of three planned trajectories, all were collision-free, with mean joint tracking error 0.018 ± 0.004 rad and end-effector deviation 4.2 ± 1.1 mm.

D. Ablation Study

We isolate the contribution of each component across representative 2-D, 3-D, 6-D, and 14-D environments (Table III): (i) *RIT* (full)*, (ii) *w/o Riemannian sampling* (Euclidean informed set), (iii) *w/o cascading* (full edge evaluation), and (iv) *w/o CARM* (oracle metric). The oracle variant uses the ground-truth metric and serves as an upper bound rather than a directly comparable method.

Observations: Low-D: Riemannian sampling provides the dominant benefit under anisotropy (+8.7% in 3-D without it; negligible in isotropic 2-D, where removing it is marginally lower, 0.916 vs. 0.921 – expected, since with $G = I_d$ the whitening reduces to the identity and only adds numerical overhead; the 0.5% gap lies within trial-to-trial spread ± 0.007). CARM closely tracks the oracle in every environment ($< 1.1\%$ overhead), with the largest residuals in 2-D and 3-D (+0.9%, +1.0%), where the cost field is learned from $G = I_d$. *High-D:* gains are smaller, and performance is primarily governed by informed-set geometry; CARM provides incremental refinement over the base metric. *Cascading* has $< 0.3\%$ impact on cost, confirming it is primarily a computational optimisation. These trends are consistent with the analysis in Sec. V-E.

E. CARM Analysis

We evaluate how well CARM approximates the oracle (ground-truth) metric by comparing RIT* under learned versus oracle metrics (Table IV). In 6-D/14-D the oracle is the inertia

TABLE IV: CARM quality analysis: mean path cost (oracle vs. learned metric). Overhead denotes relative cost increase w.r.t. the oracle; lower is better.

Env	Oracle	CARM	Overhead
2-D	0.913 $\pm$ 0.003	0.921 $\pm$ 0.007	+0.9%
3-D	2.174 $\pm$ 0.033	2.196 $\pm$ 0.027	+1.0%
6-D	1.484 $\pm$ 0.086	1.489 $\pm$ 0.082	+0.3%
14-D	8.586 $\pm$ 0.012	8.638 $\pm$ 0.144	+0.6%

matrix $M(q)$ and CARM applies a learned proximity scaling to this same $M(q)$, so the small overhead ($< 1\%$) partly reflects the strength of the prior. The low-dimensional case is therefore most informative: starting from $G = I_d$, CARM must reconstruct the obstacle-proximity cost entirely from collision feedback, yet attains overheads of only +0.9% (2-D) and +1.0% (3-D). Across all four settings CARM stays within $\sim 1\%$ of the oracle, indicating that the learned scalar field recovers nearly all the obstacle-proximity information available from collisions, an appropriate scope since collisions carry positional but not directional information.

Two additional checks support these conclusions: (a) a $3 \times 3 \times 3$ sweep over $\sigma \in \{0.05, 0.1, 0.2\}$, $\alpha \in \{5, 10, 20\}$, $K_c \in \{10, 15, 30\}$ on the 3-D Diagonal benchmark yields median costs in [2.18, 2.24] across the 27 configurations, a worst-case variation of $\pm 1.4\%$ around our default (0.1, 10, 15), which is also the best cell, indicating low hyperparameter sensitivity; (b) re-running 6-D UR10 with $G = I_d$ (no inertia prior), RIT*+CARM attains median cost 1.451 vs. 1.414 for the oracle (+2.6%) and 1.520 without CARM (+7.5%), so CARM recovers $\sim 2/3$ of the oracle gap from collisions alone. This last study also indicates that CARM is not tied to the Riemannian machinery of RIT*: since it simply rescales the cost field used for edge evaluation, it can be attached to any planner that evaluates edge costs. With an isotropic base $G_0 = I_d$, it effectively turns a Euclidean planner into a cost-space variant, and the $G = I_d$ result above suggests that similar gains may transfer.

VI. DISCUSSION AND LIMITATIONS

Discussion: The benefit of RIT* increases with both anisotropy (κ) and dimensionality. For isotropic costs ($\kappa \approx 1$), $I_R \approx I_E$ and the gap to informed-set baselines is small (within 7.4%), although weaker informed-sampling strategies (APT*) trail by up to 14.3%. As anisotropy grows, I_R aligns progressively better with the cost structure: +13.0% over BIT* in 3-D ($\kappa \approx 6$), and in 14-D ($\kappa \approx 18$) the advantage is largest at +24.8-63.5% over all five baselines. Overall, the gain scales with the tightening of the informed set under the cost geometry, while RIT* maintains competitive convergence speed (Fig. 6) despite the additional metric computations.

Limitations: Some limitations suggest directions for improvement. (i) The metric cache scales as res^d , which can become prohibitive in high dimensions; sparse, adaptive, or learnt representations could reduce this. (ii) CARM employs isotropic Gaussian kernels, which may blur directional structure near obstacles; contact-normal directions returned by the collision checker could provide the missing directional information, enabling anisotropic kernels or a tensor-valued (rather than conformal) refinement aligned with local obstacle

geometry. (iii) CARM relies on collision feedback, and its effectiveness depends on the informativeness of sampled collisions; in simpler settings recovery may be slower, motivating exploration that actively probes uncertain regions. (iv) The computational overhead of metric evaluation and caching may impact scalability in very high dimensions.

VII. CONCLUSIONS

We introduced RIT*, a Riemannian batch-informed planner that replaces Euclidean primitives with cost-consistent counterparts and learns a collision-adaptive cost refinement online via CARM. From 2-D to 14-D, RIT* produces lower-cost solutions under anisotropic metrics while remaining competitive in isotropic settings, with gains scaling with anisotropy and dimension (+13.0% over BIT* in 3-D, up to +24.0% in 6-D, and +24.8-63.5% in 14-D bimanual planning), and CARM recovers most of the oracle-metric advantage from collision feedback alone.

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